

9.2.6 Aim to be achieved in a non-technical field included in the formulation of the technical problem

In T 641/00 (OJ 2003, 352) the board considered the formulation of the technical problem in the case of mixed-type inventions. Although the technical problem to be solved should not be formulated to contain pointers to the solution or partially anticipate it, merely because some feature appeared in the claim did not mean it was automatically excluded from appearing in the formulation of the problem. In particular where the claim referred to an aim to be achieved in a non-technical field, this aim might legitimately appear in the formulation of the problem as part of the framework of the technical problem which was to be solved in order to avoid taking into account a non-technical contribution when assessing inventive step. The board referred to its decision T 1053/98, where it had considered it necessary to formulate the technical problem in such a way that "there was no possibility of an inventive step being involved by purely non-technical features". Such a formulation of the problem could refer to the non-technical aspect of the invention as a given framework within which the technical problem was posed. Thus, the board allowed an **aim to be achieved** in a non-technical field to appear in the formulation of the problem as part of the framework of the technical problem that is to be solved, in particular as a constraint that is to be met (see e.g. T 2063/09).

In T 154/04 (OJ 2008, 46) the board noted as follows: For the purpose of the problem-solution approach, the problem must be a technical one which the skilled person in the particular technical field might be asked to solve at the priority date. (as similarly held in T 641/00). The technical problem may be formulated using an aim to be achieved in a non-technical field, and which is thus not part of the technical contribution provided by the invention to the prior art. This may be done in particular to define a constraint that has to be met (even if the aim to be achieved stems from an a posteriori knowledge of the invention (as similarly held in T 641/00)). The definition of the technical problem is difficult if the actual novel and creative concept making up the core of the claimed invention resides in the realm outside any technological field, as is frequently the case with computer-implemented inventions. Defining the problem without referring to this non-technical part of the invention, if at all possible, would generally result either in an unintelligible vestigial definition, or in a contrived statement that does not adequately reflect the real technical contribution provided to the prior art.

In T 1284/04 the board held that the Comvik approach does not consider the non-technical constraints as belonging to the prior art, but rather as belonging to the conception or **motivation phase** normally preceding an invention, since they may lead to a technical problem without contributing to its solution. Such aspects have never been taken into account in assessing inventive step, irrespective of whether or not they were known from the prior art (see e.g. T 958/03 and T 506/06).

In T 144/11 the board stated that what the board had held in T 1463/11 demonstrated that a careful analysis of which parts of a claimed feature involve a business requirement could help to resolve the grey area between technical and non-technical features. A corollary of this approach, and what is seen in practice, is that a problem of the type "implement the business requirement" will normally never lead to an allowable claim. Either the

implementation will be obvious or have no technical effect, or if not, the implementation will have a technical effect that can be used to reformulate the problem essentially to "achieve the effect of the implementation". However, the implementation-type problem is just a starting point that might have to be modified when the implementation is considered. It helps when a technical problem is not apparent at the outset. Examining the business requirements like that and correctly establishing what is to be implemented ensures that all technical matter arising from the idea of the invention and its implementation is taken into account for inventive step.

In T 232/14 the board followed T 144/11 and held that using ranges of unit identifiers to label a number of (consecutive) unit identifiers of manufactured items was, at the level of generality at which it was claimed, on the business side of the line between technical and non-technical subject-matter. The ranges of unit identifiers did have a meaning for the business person. They corresponded to batches of units produced on a production line. Even if the "determining of ranges of unit identifiers" achieved a technical effect, such as reducing data storage and data bandwidth requirements, it was a matter of routine design for the skilled person, a software programmer or a database expert, based on common general knowledge to store the first and the last element of a list of items, instead of the whole list.

In T 2052/12 the invention dealt with parking meters that could credit money on smart cards, but also stored a repayment limit on the smart card to prevent theft. The board held that the claimed method involved technical features carried out by the data carrier, in particular the storing, setting and reducing of the repayment limit, the checking of the reimbursement value and its contingent addition to the memory value, and the checking of key information identifying the communication facility's credit authorization. It held that these technical features brought about technical effects beyond the mere implementation of the claimed method in such a way as to achieve the aims corresponding to the non-technical features.

In T 581/14 the board took the view that a reduction in complexity of calculation could be a technical aim.

In T 2314/16 the invention concerned the distribution of rewards to participants in an affiliate marketing scheme, by which an influencer (called user in claim 1) received a reward for advertising a product or service on a blog or on social media. Participating influencers were each allocated a portion of an advertisement banner displayed on a web site. The visitors to the website did not see the user areas, only an advertising banner. When the visitor clicked on the banner, the user whose portion was clicked on got a reward, distributed according to the sizes of the image portions. The partial areas were allocated such that the reward distribution rates matched the degree of contribution of each user to the advertising. The board held that the specification of the business method ended with how to determine the reward distribution ratio. The features of dividing the advertisement display area into partial areas and allocating each partial area to a user such that when the partial area was clicked on the user got a reward, were based on technical considerations of the web page system. It was not motivated by any business considerations. In order to come up with this idea, one needed to understand how a web

site is built, and in particular how an image map works. Thus, this feature could not be part of the non-technical requirements. Instead it was part of the solution that had to be evaluated for obviousness.

In T 550/14 the board declined to define criteria that the examining division should use to prove that a feature is not technical, in keeping with previous case law. However, as stated in T 2314/16, over the years the case law has provided guidance on the issue of technicality, most recently the framework for discussion given in T 1463/11 to help classify whether borderline features of a claim are on the technical or the non-technical side. The board gave one piece of advice for examining divisions: where possible to search for and start from a document that already discloses some of the alleged non-technical features, thus avoiding the discussion for these features (see e.g. T 756/06 or T 368/05).

9.2.7 The notional business person

In T 1463/11 the board introduced the notional business person. The board noted that if the essential idea of the invention lies in a non-technical field (usually one excluded by Art. 52(2) EPC, such as business, programs, or presentations of information), the objective technical problem often amounts to a statement of requirements that any implementation must meet. The assessment of what is and what is not technical is, therefore, a critical step in the formulation of the objective technical problem. A non-obvious difference over the prior art leads to a positive outcome, if it is deemed technical; but a non-obvious difference that is deemed non-technical leads to a negative outcome. This often leads to opposing definitions of the problem and must therefore be analysed precisely. The formulation of the objective technical problem in terms of non-technical requirements raises the question of what requirements the business person (for example) can actually give to the technically skilled person. Naturally, any requirement that is purely a business matter can be included. However, in the assessment of inventive step, the business person is just as fictional as the skilled person of Art. 56 EPC. The notion of the skilled person is an artificial one; that is the price paid for an objective assessment. So it is too with the business person, who represents an abstraction or shorthand for a separation of business considerations from technical. A real business person, a real technically-skilled person, or a real inventor does not hold such considerations separately from one another. Thus, the notional business person might not do things an actual business person would. He would not require the use of the internet. This approach ensures that, in line with the Comvik principle, all the technical matter, including known or even notorious matter, is considered for obviousness and can contribute to inventive step. Similarly, the notional business person might do things that a real business person would not, such as include requirements that go against business thinking at the time - a sort of business prejudice. If this were not the case, business requirements would need to be evaluated and would contribute to inventive step, contrary to the Comvik principle.

In T 144/11 the board followed T 1463/11 and held that the technical skilled person must receive a complete description of the business requirement, or else he would not be able to implement it and should not be providing any input in the non-technical domain.

In T.1082/13 the board held that the notional business person, as interpreted within the framework of T.641/00 knows all about the business related requirements specification and knows about the fact that such business related concepts can be implemented on a computer system. The choice of where to do a calculation in a distributed system is not necessarily technical, but can also be driven by administrative considerations. What the notional business person does not know, however, is how exactly it can be implemented on a computer system. This is in the sphere of the technical expert and subject to the assessment of inventive step. When referring to prejudices, it has to be carefully analysed, whether it is actually a technical prejudice or, in fact, a business prejudice.

In T.1408/18 the board found that a business person who wanted to offer a product enabling users to make a transaction on a single terminal device would specify that the transaction could go ahead only after user authorisation and that, in keeping with the latest trends, it would be desirable for users to be able to make all necessary entries on their smartphone. By contrast, using a TAN-based process, including dealing with how the TAN could be transmitted securely, fell within the sphere of a technical expert. Building on traditional PIN-based password authentication, using a TAN, i.e. a one-time password, added a second layer of security. The associated interaction of two applications and communication channels for obtaining and providing a TAN led to two-factor authentication guaranteeing greater security. So, regardless of how a TAN-based process was used in practice, there were technical considerations behind it that went beyond the technical understanding that could be expected of a business person.

In T.2455/13 the board endorsed the above reasoning in T.1082/13 and added that, features were not technical where they were specified merely as modules on an abstract meta level and represented functions the "non-technical skilled person" would base their idea on; after all, in outlining these functions, that person would not be considered to have dictated any technical features either. It was only once actual implementation steps had been specified in the claim that such modules could be treated as technical features.

In T.737/14 the board held that the proper application of T.641/00 required a thorough analysis of the business constraints when formulating the problem to be solved before investigating what the skilled person would have done to solve it.

In T.817/16 the board considered that a useful test for determining whether such technical considerations were present was to ask whether the non-technical features would have been formulated by a technical person rather than by a non-technical person or persons (T.1214/09, T.1321/11, T.1463/11, T.136/13). This was not an enquiry into the actual state of technical or non-technical knowledge at the effective filing date; the question was rather whether the knowledge required for coming up with the non-technical features in the particular case was of a kind that only a technical person, i.e. a person not working exclusively in areas falling under Art. 52(2) EPC, could possess.

In T.1902/13 the board held that a business consultant, who wanted to assess the competence of an organisation, would design a set of rules and questions which could be reused for another organisation. Automating parts of this process did not render it technical. The board did not agree that the business person and the skilled programmer

would have to sit together in order to elaborate a workable solution. Rather, the underlying administrative concept would be provided to the programmer as a requirements specification.

In T.1749/14 the board held that the notional business person might come up with the abstract idea of avoiding the customer having to provide PIN and account information to the merchant. The invention however required a new infrastructure, new devices and a new protocol involving technical considerations linked to modified devices and their capabilities as well as security relevant modifications of the transfer of sensitive information using new possibilities achieved by the modifications to the previously known mobile POS infrastructure. This went beyond what the notional business person knew and concerned technical implementation details which were more than a straight-forward 1:1 programming of an abstract business idea. This was in the sphere of the technical expert and subject to the assessment of inventive step (see T.1082/13).

9.2.8 Credible technical effects

In T.258/97 the board held that an assessment of inventive step could only be based on those elements and aspects of the invention in respect of which a technical effect could be established (see also T.1461/12). Whether an invention caused a technical effect was essentially a question of fact. While the EPO has a duty to determine such facts in examination proceedings, the onus is upon the applicant to cooperate in said determination, in particular in the event of doubt (see also T.953/04, T.1958/13).

In T.27/97 the board held that an abstract algorithm was relevant to inventive step only if a technical effect could be established which was causally linked to the algorithm, such that the technical effect provided a contribution to the solution of a technical problem and thereby conferred a "technical character" on the algorithm.

In T.953/04 the board followed decision T.258/97 in this respect and noted that if the doubts about the technical character of the invention could not be resolved, the features in question had to be ignored in assessing inventive step (see also T.27/97).

In T.823/07 the board, following decision T.1143/06, stated that the manner how cognitive content is presented to the user may only contribute to the technical solution of a technical problem if the manner of presentation (exceptionally) shows a **credible technical effect** (see also T.1575/07, T.1741/08, T.1562/11).

In T.1958/13 the invention was concerned with copying and pasting on a touchscreen display by means of a "single-drag gesture". Although the board deemed those effects, in principle, to be technical effects, since in the end they **aim at providing tools** which serve or assist user activities (see e.g. T.643/00), the question whether they were actually achieved depended exclusively on **subjective user skills or preferences**. Therefore, the board was not satisfied that they could be regarded as objectively credible technical effects for the purpose of formulating the objective problem to be solved (see also T.1567/05, T.1841/06 and T.407/11; see also T.77/14 on gesture-based functions).